SMART ENERGY MANAGEMENT SOFTWARE

WORKING EXPLANATION-WEB AND ANDROID APPLICATION

DOCUMENT PURPOSE: A simple explanation of how the completed system is structured and how its main parts work together.

Application Types: Web application and Android mobile application.

Core Technology: Java, Spring Boot, Rest API and MySQL, Html/CSS for web application and Android Studio for app application.

1.ProjectOverview:

Smart Energy Management Software (SEMS) is designed to help an organization monitor and manage its electricity consumption. It brings energy information into one system so users can review usage, compare consumption, understand electricity expenses, and identify areas where energy may be wasted.



The project has two user-facing applications: A website that can be accessed through a browser and an Android
application that can be accessed from a mobile device. Both applications are intended to use the same backend
services and central database when configured to connect to the same server.

2. Overall System Architecture:

The system can be understood as three main layers:

1. Presentation layer: The website and Android application show screens, accept user actions and display information.

2. Application layer: The Spring Boot backend receives requests, checks and processes the information, applies business rules and sends responses.

3. Data layer: The MySQL database stores and retrieves the information needed by the application.

Basic request flow: User → Web/Android screen → REST API request → Spring Boot processing → MySQL
read/write → API response → Screen update.

The website and Android app do not need separate databases. If both are configured with the same backend API
and database, they can display and update the same stored information.

3. How the Website Works:

A user opens the website in a browser and navigates through the available pages. When the user selects a feature or submits a form, the web page sends a request to the backend API. Spring Boot processes the request and, when required, reads data from or writes data to MySQL. The backend then returns a response, and the website displays the result.

Typical website flow:

- Open the application URL in a browser.
- Enter login details on the login page, if login is enabled.

• After successful authentication, open the dashboard or a required module.

• View information, submit forms, or perform available add/edit/delete actions.
• The browser sends the relevant request to Spring Boot.
• The page displays the response received from the backend.

4. How the Android Application Works

The Android application provides a mobile interface for the system. It collects user actions through mobile screens and communicates with the Spring Boot backend through REST API calls. The backend handles the request and accesses MySQL when data needs to be retrieved or saved. The response is then displayed in the Android application.

Typical Android flow:

• Open the application on an Android device.
• Sign in, if authentication is enabled.
• Navigate to the available dashboard and modules.
• Request or submit information through the app.
• Receive the backend response and show the result on the mobile screen

5. Login and User Access

The login screen is the entry point for users. In a typical login process, the application sends the entered credentials to the backend over an API. The backend validates the credentials against the configured user records and returns an appropriate result. Access to protected pages or APIs should be restricted according to the authentication and authorization features actually implemented in the project.



6. How the Main Modules Work:

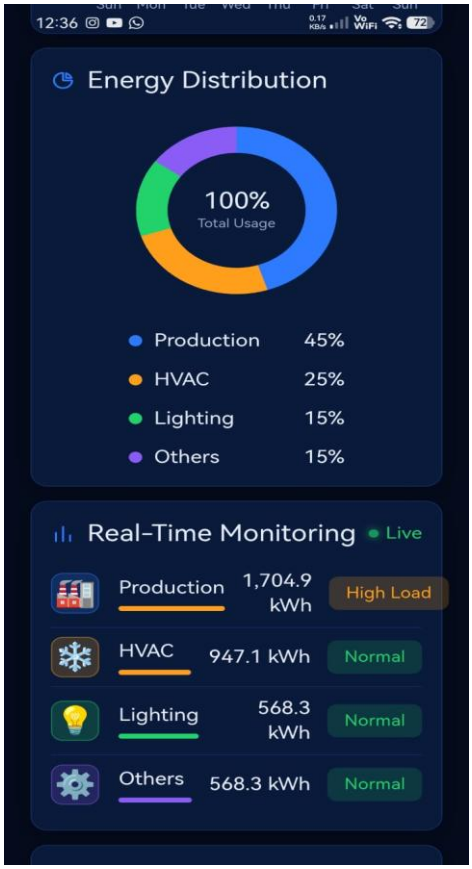
The following descriptions explain the purpose of the main modules. The exact operations available in each module depend on
the screens, API endpoints and database functions implemented in your application.

1.Dashboard Management:

Dashboard Management helps to Displays a summary of key energy information and shortcuts to important sections.


Dashboard
Energy Monitor
Smart Energy Management Software
Monitor · Optimize · Reduce Carbon Footprint
Live
Thu, Nov 05, 2026 12:36 PM
Total Energy Usage
82,915 kWh
↑ 40.8% vs last 30d
Energy Cost
₹ 6,63,317
↑ 40.8% vs last 30d
Energy Savings (potential)
7.6%
≈ ₹ 50,244 / month possible
Carbon Footprint


Carbon Footprint
35.1 tCO2
↑ 43.6% vs last 30d
Energy Consumption Overview
1D
1W
1M
3M
1Y
Energy Usage (kWh)
Target Usage
3,980.8 kWh
Sat · peak
Sun Mon Tue Wed Thu Fri Sat Sun
Energy Distribution
100%
Total Usage


Energy Distribution
100%
Total Usage
Production 45%
HVAC 25%
Lighting 15%
Others 15%
Real-Time Monitoring
Live
Production 1,704.9 kWh
High Load
HVAC 947.1 kWh
Normal
Lighting 568.3 kWh
Normal
Others 568.3 kWh
Normal


Recent Alerts
View All →
High energy consumption
MTR-001 · 481 kW (120.2 kWh in 15 min)
01 Nov 13:45
Power factor low
MTR-001 · 0.69
13 Oct 08:30
Usage anomaly detected
MTR-001 · 63.4 kWh vs usual 8.3 kWh for this hour
30 Oct 21:00
Peak demand this month
MTR-001 · 558 kW
31 Oct 09:00
Energy Saving Tips
Maximum-load intervals use 51% of your energy. Shifting part of that load to off-peak hours can cut peak demand and cost.
View Recommendations →

2.Real-Time Energy Monitoring:

Presents the latest available electricity readings. Data is real-time only when a live meter/device feed is connected otherwise, the screen may show stored, imported or manually entered readings.



3.Building and Department Management:

Organizes the organization into buildings and departments. Where implemented, users can add, view, edit or delete building and department records.



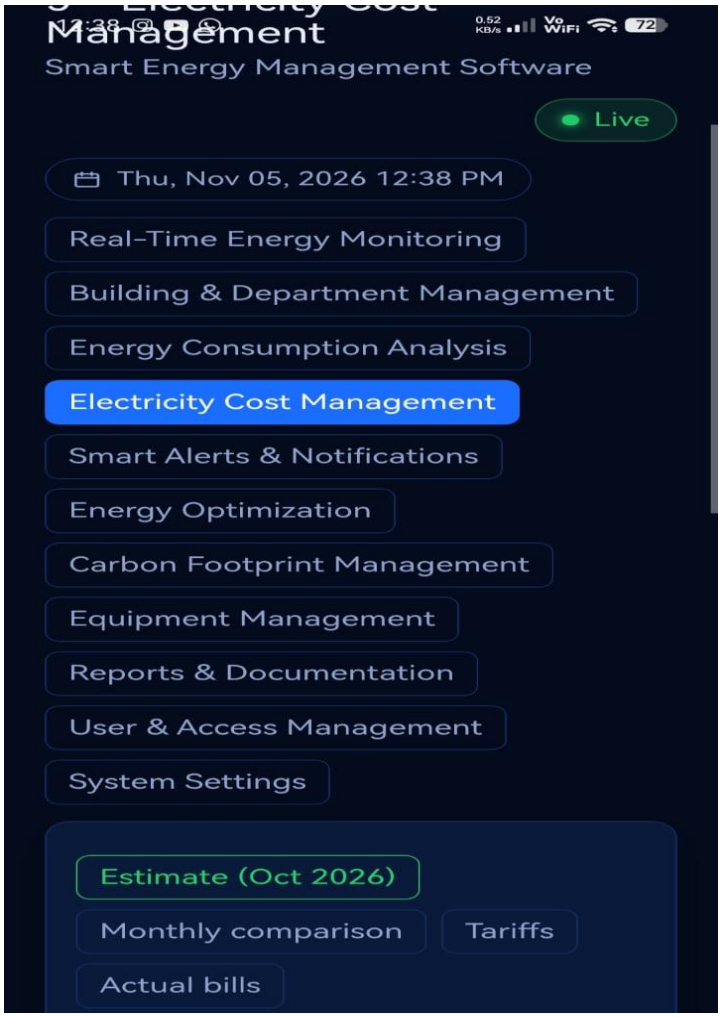
4.Energy Consumption:

Analysis Helps review consumption over selected periods and compare usage patterns using available readings.



5.Electricity Cost Management:

Calculates or displays electricity expenses using the energy units and tariff/rate values configured in the system.



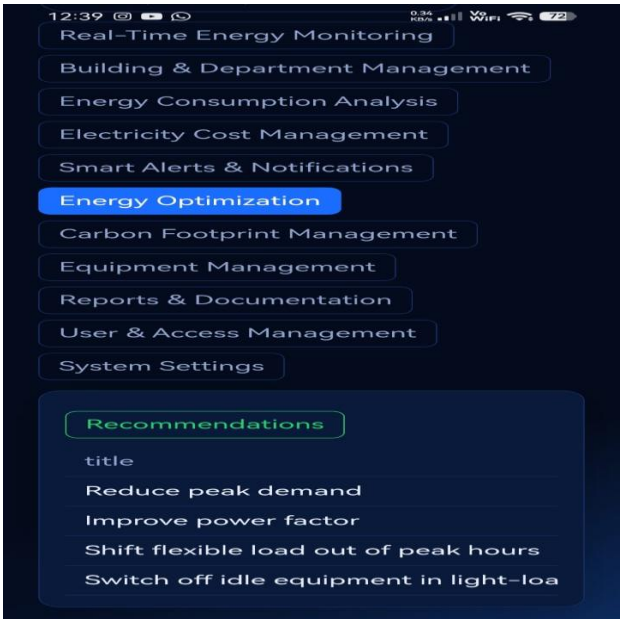
6.Smart Alerts and Notifications:

Shows warnings or notifications when configured conditions are met, such as an unusual reading or a usage limit being crossed.



7.Energy Optimization:

Helps users identify possible opportunities to reduce unnecessary electricity consumption based on the available usage information.



8.Carbon Footprint Management:

Estimates emissions from electricity consumption using a configured emission factor, if that calculation is implemented.



9.Equipment Management:

Maintains information about equipment or electrical assets and may associate them with buildings or departments.



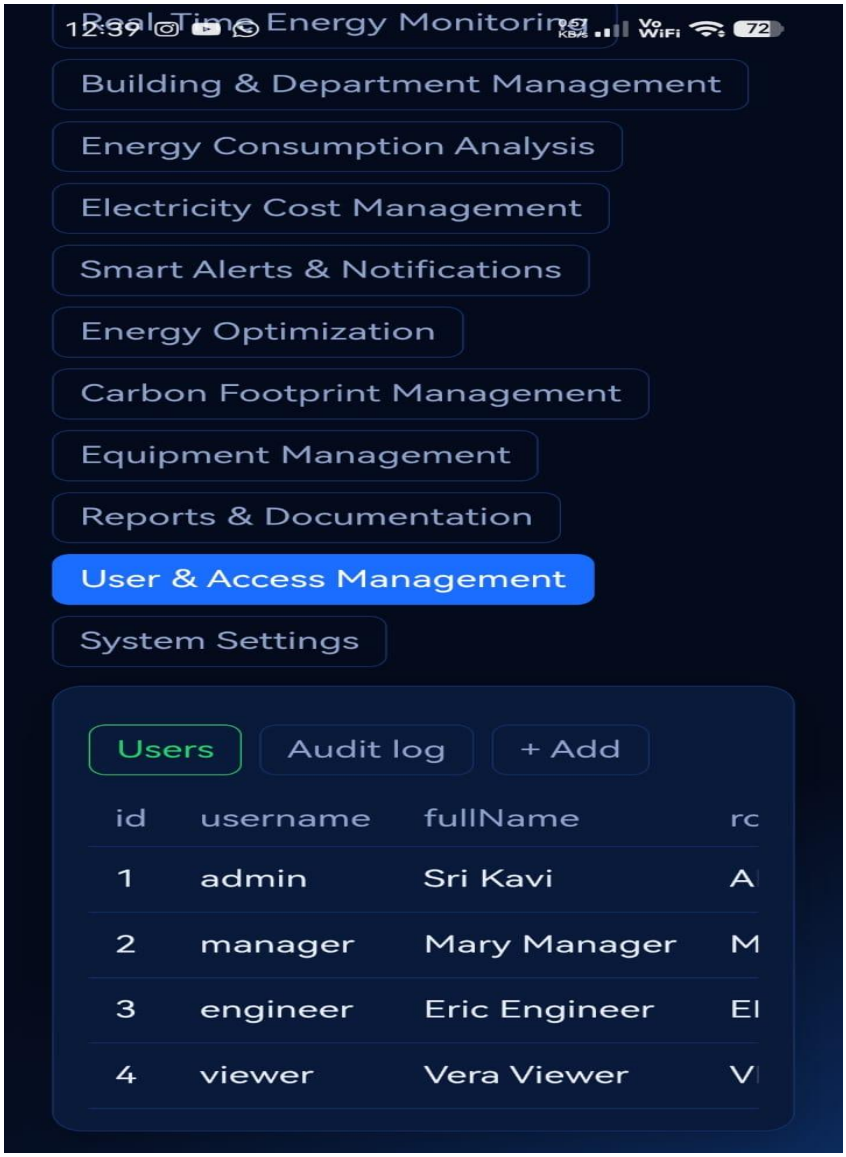
10.Reports and Documentation:

Presents or exports energy usage, cost or efficiency information in the formats supported by the application.



11.User and Access Management:

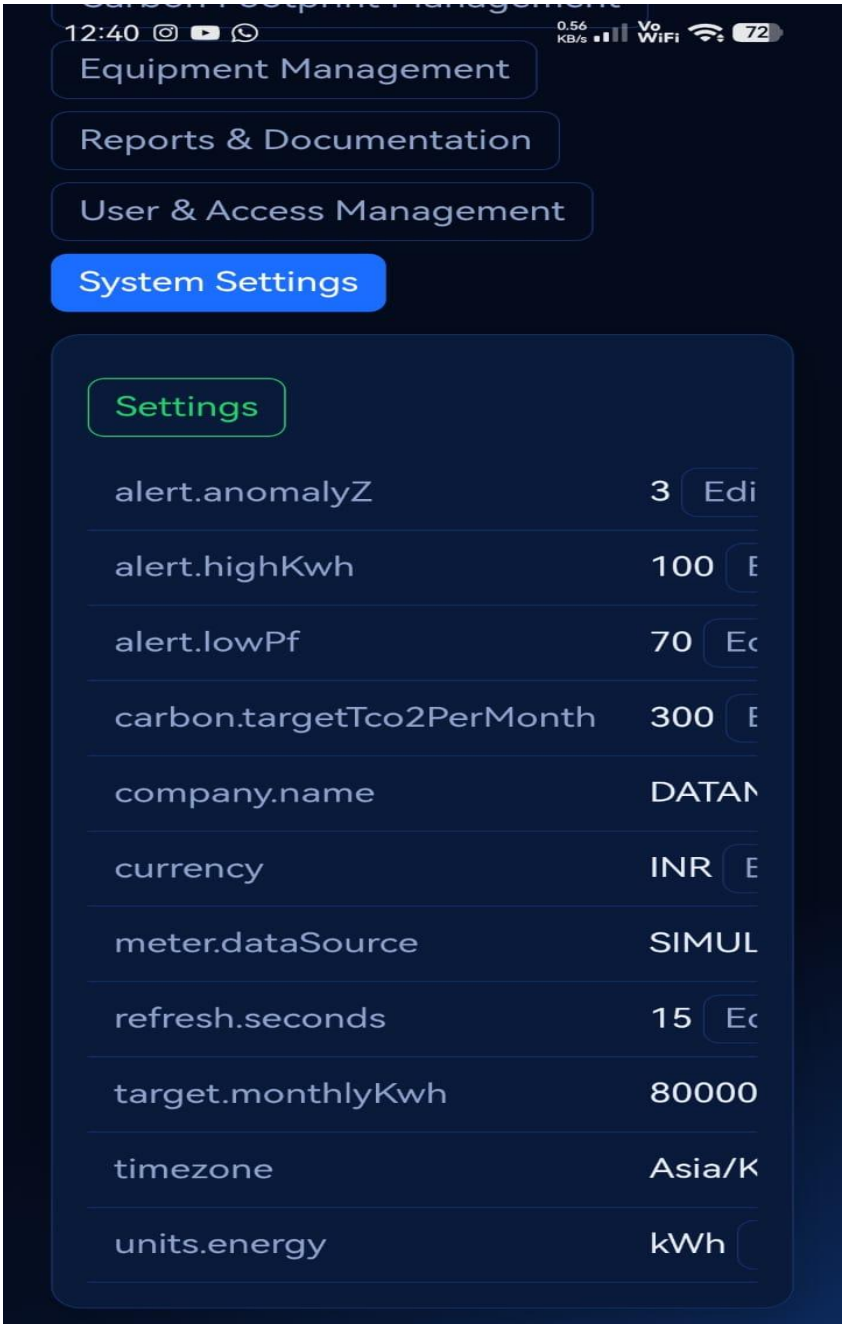
Manages user records and access permissions where these functions are implemented.



12.System Settings:

Stores configurable application values such as tariffs, thresholds, preferences

Or other settings supported by the system.


12:40
0.56 KB/s
Vo WiFi
72
Equipment Management
Reports & Documentation
User & Access Management
System Settings
Settings
alert.anomalyZ 3 Edi
alert.highKwh 100 E
alert.lowPf 70 Ec
carbon.targetTco2PerMonth 300 E
company.name DATAN
currency INR E
meter.dataSource SIMUL
refresh.seconds 15 Ec
target.monthlyKwh 80000
timezone Asia/K
units.energy kWh